

Motherboards				
Boards Above Rs. 10,000 and Up To Rs. 18,500				
Brand Model No.	ASUS Rampage Formula	ASUS P5Q Premium	Gigabyte P45-DS4P	ASUS Striker 2 Formula
				
Chipset	Intel X48	Intel P45	Intel P45	NVIDIA nForce 780i
Socket	LGA 775	LGA 775	LGA 775	LGA 775
Northbridge / Southbridge	X48 / ICH9R	P45 / ICH10R	P45 / ICH10R	nForce 780i SLI / nForce 570 SLI
Price (Rupees)	17000	17750	17890	18350
Plus (+)				
Minus (-)				
Grand Total (Out of 100)	74.79	75.40	72.87	75.35
Features (out of 60)	39.83	40.44	37.64	41.27
Performance (Out of 40)	34.96	34.96	35.24	34.09
Front Side Bus Speed (MHz)	1333 MHz	1333 MHz	1333 MHz	1333 MHz
Memory Type	DDR2	DDR2	DDR2	DDR2
Max Memory Supported (GB) / No of DIMMs	8 GB / 4	8 GB / 4	8 GB / 4	8 GB / 4
No. of SATA / IDE Ports	6 / 1	10 / 1	6 / 1	6 / 1
E-SATA Support / No of E-SATA Ports	✗ / NA	✗ / NA	✗ / NA	✓ / 2
No of PCI Express graphics slots	2	4	2	3
No. of PCI / PCI x1 / PCI x4 Slots	2 / 3 / ✗	2 / 1 / ✗	1 / 3 / 1	2 / 2 / ✗
Multi GPU Support (SLI/CrossFire/N) / No of PCIe Lanes	Y / Crossfire x16	Y / Crossfire x8	Y / Crossfire x8	Y / 3 way SLI, (x16, x16, x8)
Integrated Audio Controller	ADI 1988B	ADI 1988B	Realtek ALC 889A	ADI 1988B
SPDIF / Optical Port	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
No of LAN Ports / 10/100 / Gigabit	2 / Gigabit	4 / Gigabit, 10/100	2 / Gigabit	2 / Gigabit
Integrated Wi-Fi (Y/N)	✗	✗	✗	✗
No. of rear USB / FireWire Ports	6 / 1	10 / 0	8 / 2	6 / 1
All Solid State Caps (Y/N)	✓	✓	✓	✓
BIOS options				
Overclocking / Overvolting (So 10)	9.25 / 9.5	9 / 9	8.25 / 8	9.5 / 9.25
CPU voltage range (volts)	1.1 to 2.4v	0.85 to 2.1v	0.5 to 2.3v	1.1v to 2.2v
Memory voltage range (volts)	1.8 to 3.4v	1.8 to 3.08v	1.8 to 3.0v	1.8 to 3.4v
Package Bundle				
No of SATA / PATA Cables Provided	6 / 1	6 / 1	6 / 1	6 / 1
CD's	Drivers, STALKER game, 3D Mark 06 Advanced Edition, Kaspersky Antivirus	Drivers	Drivers	Drivers, STALKER game, 3D Mark 06 Advanced Edition, Kaspersky Antivirus
Other Miscellaneous Accessories	LCD poster, USB and FireWire bracket, optional fan	USB and FireWire bracket	USB and FireWire bracket	LCD poster, USB and FireWire bracket, optional fan
Performance				
PC Mark 2005:				
CPU / Memory	9689 / 6552	9695 / 6615	9652 / 6453	9625 / 6912
Graphics / HDD	18293 / 7147	18366 / 7014	18462 / 7072	18286 / 6829
Overall	10284	10494	10476	10436
SiSoft Sandra 2008:				
CPU Arithmetic (Dhrystone ALU / Whetstone iSSE3)	55616 / 44264	55677 / 44051	55477 / 43994	55489 / 44019
CPU Multimedia (Integer / Floating)	330318 / 181063	330650 / 181126	329438 / 180393	328265 / 180298
Drive Index (MB/s) / Random Access Time (ms)	73 / 6	72 / 6	72 / 6	67 / 13
SiSoft Memory bandwidth score (Integer / Floating)	6787 / 6780	6829 / 6829	6526 / 6576	6587 / 6582
3D Mark 2006 Marks / CPU Score	16829 / 4999	16870 / 4951	16562 / 4827	16627 / 4926
CINEBENCH R10 (CPU)	11366	11849	11856	11714
WinRAR 3.8	2149	2213	2193	1628
Video Encoding (DivX 6.8) Time (sec)	113.4	112.8	113.7	114.3
File Transfer Test (4 GB File) (MBps)	135.8	144.1	138.4	147.1
Doom 3 FPS (1600x1200, 4xAA, 4xAF / 640 x 480, low detail)	211.5 / 294.9	212 / 295.7	206.4 / 297.5	206.3 / 288.3
S.T.A.L.K.E.R Shadow Of Chernobyl (1680x1050, 4xAA, 4xAF / 800 x 600, low detail)	179.53 / 679.69	181.07 / 676.24	180.43 / 891.4	180.74 / 892.5
Prey (1680 x 1050, 4x AA, 4x AF, 640 x 480, low detail)	197.1 / 354.5	210.5 / 356.8	206.6 / 349.2	209.5 / 346.2

tions are some of the most attractive around. The ASUS M3A79-T Deluxe was unique by supporting 4-way Crossfire or as ATI calls it, Quad Crossfire. Intel’s own DX48BT2 was also present but this board pales in comparison to the two other X48 boards in

test. It’s somewhat plain-jane looking with a black PCB and Intel’s traditional all aluminium heatsinks on both chipsets. Both MSI’s boards, the X48c Platinum and the P45 Platinum were good lookers although somehow we do not care for the elaborate cir-

cular shaped heatpipe solution atop the former which looks overly loud and garish. The X48c Platinum is very well laid out; the PCI slots can be populated even with a Crossfire setup – this is rare. The SATA ports are placed away from the primary PCIe slot

and even a full length, double-slot card will not render them unusable. Equally important for some is the support for both DDR2 and DD3 memory on the X48c; this board has six memory DIMMs; two for DDR2 and four for DDR3. This means you can upgrade to DDR3 later, when it’s more affordable. All these features come with a caveat – there are only four SATA ports and this may not be enough for those running multiple HDDs and RAID arrays. The P45 Platinum is equally well laid out and the heatpipe solution is much easier on the eye; in fact the heatsinks atop the Northbridge have been designed to appear like a four cylinder engine block complete with cooling fins. However, this high-rise heatsink couple with the fact that the heatsink cooling the mosfets on the other side of the board are also quite high, means your fingers face a little abuse while fastening the CPU cooler. MSI provides eight SATA ports on the P45 Platinum; as if to make up for the X48c Platinums’ deficiency in this regard.

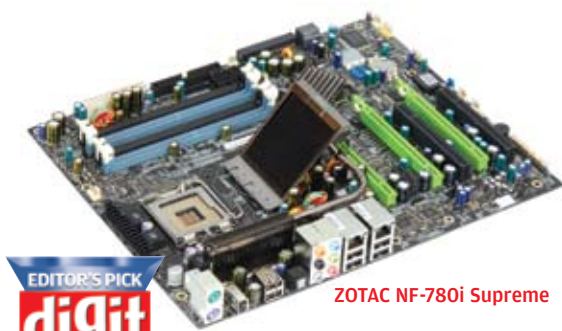
The other three 780i SLI boards were from EVGA, XFX and ZOTAC and these boards look identical since these vendors choose to use NVIDIA’s standard PCB; in fact these PCBs are even made by NVIDIA so the finest details are identical too. EVGA has the most SATA connectors bundled and these are an attractive black in colour. The PATA connector too is encased in a round bracket to aid airflow – thoughtful. ZOTAC bundles a WiFi kit with their motherboard which is the only difference between these boards aside from their bundles and prices.

The ASUS boards have the most feature rich BIOSes in terms of overclocking options and they are closely followed by Gigabyte. Never before have we seen such minute settings and you can tweak virtually any component you choose. There are also multiple reference voltage points that you can set. Overclockers will spend a lot of time here. We also think ASUS does a very good job of making their BIOSes new-

bie friendly with help options at every stage. The settings are also very transparent and menu structure is good. Both Gigabyte and ASUS support updating their BIOS through USB drive and this is so much more convenient then hunting around for a floppy drive and working diskette. Although most vendors have programs that allow flashing the BIOS from Windows itself we feel the option of adding support for flashing from a USB drive from within the BIOS is a very useful feature.

Performance

In terms of overall performance it seems NVIDIA’s 780i SLI chipset cannot match Intel’s X48 and P45 chipsets. Of particular attention is the storage subsystem where Intel definitely has a lead and NVIDIA’s Southbridge has latency issues in terms of hard drive access and during our file copy test. The ASUS M3A79-T Deluxe trails by a good margin on account of its AMD processor but from amongst the Intel boards ASUS’ P5Q3 Deluxe was just ahead of the competition. As is the nature of motherboard testing the differences when using identical components at stock speeds is very minute; but the X48 and P45 chipsets generally outperform the 780i in both CPU and HDD based benchmarks. It also appears that DDR3 memory gives both the ASUS P5Q3 Deluxe and the Intel DX48BT2 a slight edge. Although we feel that DDR3 will really shine at speeds of 1600 MHz and above its nice to see some small performance increments being gained at relatively slower speeds of 1333 MHz and with looser timings.



Winner(s)

Our Best Buy award goes to MSI’s P45 Platinum; a highly competent board for most high-end users at a great price point of Rs. 10,800. It’s got a load of connectivity options and even supports Crossfire. It’s definitely no slouch in terms of performance and shames some costlier boards. Our overall best performer was also to be found in this category (shockingly) as ASUS’ P5Q3 Deluxe demolishes even the ultra-high-end offerings. If you want the fastest board out there, this is the one to buy. It’s very feature-rich as well, and you won’t want for connectivity options which it has a bonanza of. DDR3 support means its reasonably future proof. The price of Rs 13,850 may cause you to go “ouch” but it’s superlatively well built if that massages your sore wallet. Our Editors Pick was the full x16 SLI supporting ZOTAC NF-780i Supreme. This board has all the connectivity options, gives you the option of 3-way SLI and throws in a WLAN access point all for Rs. 12,999 – good value for money for someone building a fast gaming rig.

Motherboards above Rs. 19,000

Wallet busting superlatives

When talking of Rs. 19,000 for just one motherboard you just know that the nonsense will be left far behind. 20-odd grand is either a good price for a superb motherboard or just a ridiculous price for an atrociously expensive and utterly undeserving offering. Entire PCs can be built for less, which means these boards either have some serious plus points in terms of features or connectivity,